

KANU MADHOK

952-303-1045 • madhok.kanu@gmail.com • github.com/kmadhok • linkedin.com/in/kanu-madhok

Live Demo: [NL-to-SQL Copilot](#) — [Dockerized Streamlit on GCP Cloud Run](#)

PROFESSIONAL EXPERIENCE

Senior Data Analyst — AI/ML Engineering

Sep 2024 – Present

Walmart Data Ventures

Chicago, IL

- Designed and shipped a GenAI text-to-SQL copilot over Walmart BigQuery (RAG + 12K-query corpus, multi-agent LLM orchestration) on GCP Cloud Run; 95% SQL accuracy, ~80% faster query creation, rolling out org-wide to 500+ users.
- Productized two AI skills (cp-analytics, Simple CP Interaction) org-wide as official internal Wibey (Walmart Claude Code fork) skills after partnership review with Product and Data Science leadership — first AI skills productized across Walmart Data Ventures.
- Architected a 9-stage gated autonomous analytics pipeline (SELECT→PLAN→EXECUTE→INTERPRET→PACKAGE→CHART→NARRATE→VALIDATE→COMMIT); 146 published analyses on a 740K-panelist dataset in 6 weeks with analyst review only at publish time.
- Built jira-ticket-worker v3.0 (6-gate triage, 4-phase workflow, 10-retry self-healing, SQL similarity reusing 80%+ of past solutions); cut Jira ticket turnaround from ~60–90 min to ~5 min review-and-approve across 400+ data requests.
- Built 3 MCP servers (24+ tools) over stdio enabling agents to reason over a 43-table BigQuery dataset with zero hardcoded knowledge; cut agent latency from ~90–120s to ~20–25s.
- Productionized a 4-layer ML pipeline (7,327 LOC, mypy --strict, 131 pytest tests) reducing 652K candidate column pairs to 50 classified relationships in 45 seconds, replacing weeks of manual analyst schema-mapping effort.

AI Engineer (Consulting Capstone)

Jan – Aug 2024

FTI Consulting

Chicago, IL

- Deployed a RAG application for proposal generation adopted into the FTI workflow; cut proposal time ~30%, +34% generation quality (query rewriting), +42% retrieval (adaptive chunking). Dockerized Streamlit on AWS EC2. **Won Best in Show** — UChicago MSADS Capstone.

Graduate Student Researcher — AI & Experimentation

Jan – Sep 2024

Data Science Institute, University of Chicago

Chicago, IL

- Built a persona-driven multi-LLM (OpenAI + Gemini) chatbot with temperature/persona tuning for a staged donation experiment (500+ participants); persuaded 15% to donate to an opposing cause via controlled A/B tests.

Data Scientist

Oct 2022 – Mar 2023

Innovare (EdTech Startup)

Chicago, IL

- Built a logistic-regression student-risk model across 10+ districts (5K+ students): +7% retention, –15% course failures. Automated ELT pipeline to BigQuery, +23% reporting accuracy.

Consultant, Enterprise Platform Management (Oracle)

Sep 2021 – Sep 2022

Baker Tilly

Chicago, IL

- Delivered Oracle enterprise platform implementations end-to-end with client business and IT stakeholders; built RPA workflows automating high-volume client operations and AWS computer-vision solutions from problem definition through deployment.

TECHNICAL SKILLS

AI / Agentic: Claude Code (primary dev environment), authored + published Claude Code skills, Wibey (Walmart Claude Code fork), MCP servers, multi-agent orchestration, agentic workflows, RAG, NL→SQL, context & prompt engineering, evaluation pipelines

AI Stack: Anthropic Claude, Vertex AI Gemini (2.5 Pro/Flash/embeddings), OpenAI API, ChromaDB, FAISS, MinHash/LSH, LangChain, FastAPI, Streamlit, Pydantic

Languages / ML: Python, SQL, R, PySpark; TensorFlow, PyTorch, scikit-learn; A/B testing, time series, clustering, Bayesian methods, logistic regression

Data & Infra: BigQuery, Databricks, MySQL, Hive; Docker; GCP (Cloud Run, Vertex AI), AWS (EC2); asyncio • Power BI (DAX), Tableau, Looker Studio • RPA, AWS computer vision, Oracle enterprise platform

EDUCATION

University of Chicago — M.S., Applied Data Science

GPA: 4.0

Loyola University Chicago — B.S., Information Systems & Economics

GPA: 3.5